MULTI-AGENT ELECTION LAB · PART 2

Who steers the winner on purpose

Two deliberate manipulations, each run against an untouched twin town: who wins, how, and can anyone catch them?

Team Eclipse · Experiments 5 and 6 · towns of 100 people, same engine

THE QUESTION

If someone steers the result on purpose, does it work, and can anyone catch them?

EXPERIMENT 5

A watchdog that only sees the vote totals

An attacker slowly changes voters' minds. 10 attack styles, 420 runs.

EXPERIMENT 6

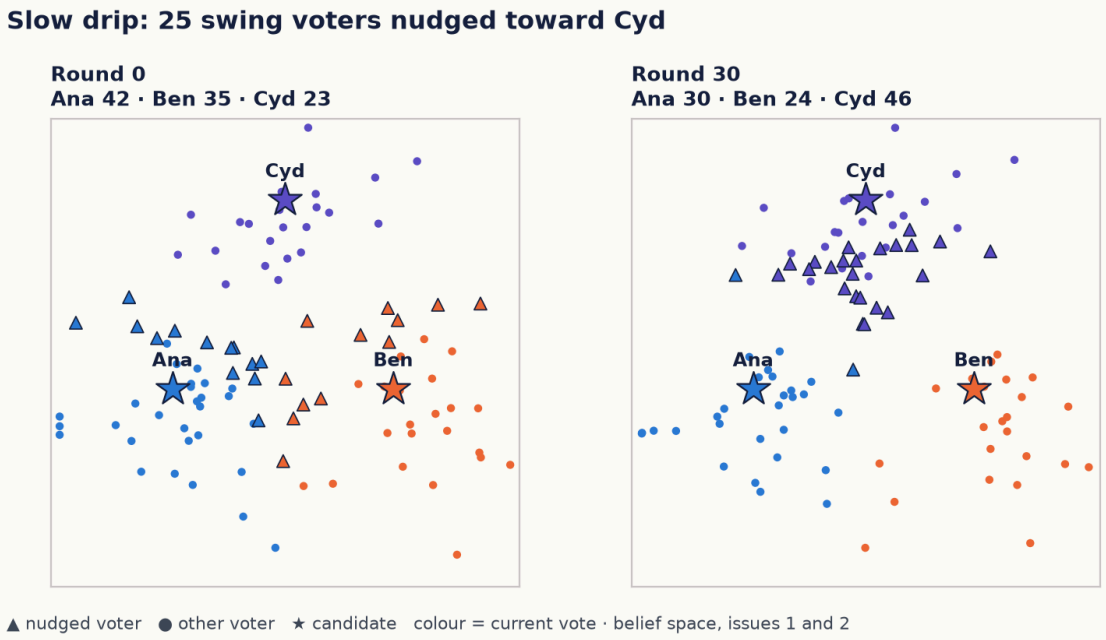
Campaigns vs echo chambers

Three parties pick a campaign playbook. All 125 combinations × 3 towns.

Every manipulated town is compared with a twin that is identical except that nobody manipulates

SETUP FOR EXPERIMENT 5

Can a watchdog catch a slow takeover?



One attack, start and end. Triangles are the voters being pushed. Over 30 rounds they drift toward Cyd, and Cyd goes from third (23 votes) to first (46).

THE TOWN

100 people, three candidates: **Ana 42, Ben 35, Cyd 23**. Everyone votes for the candidate closest to their views.

THE ATTACKER

Has a budget to nudge chosen voters' views toward Cyd, a little every round. We tried 10 styles: fast or slow, few or many voters.

THE WATCHDOG

Sees **only the vote totals**, like an election monitor. It wins if it sounds the alarm **before Cyd wins three rounds in a row**.

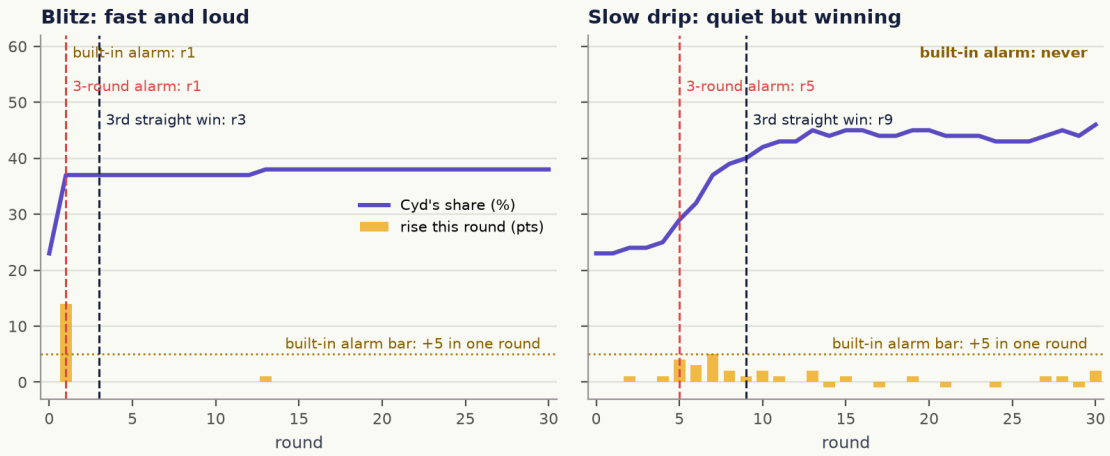
THE BUILT-IN ALARM

Fires if the target jumps more than 5 points in a single round.

Views also drift a little on their own every round · 30 rounds · every attack has an untouched twin town

EXPERIMENT 5

Watch the trend, not the jump



Purple: Cyd's vote share. **Yellow:** how much it rose that round. The blitz jumps at once; the slow drip never rises more than 5 in a round, yet wins by round 9.

WHAT WE GOT

The one-round alarm caught **20 of 25** winning attacks: it misses slow, patient ones. Our best rule, **"alarm if the target rose more than 4 points over 3 rounds"**, caught **25 of 25**, with **no false alarms** in 40 untouched towns.

CAN THE ATTACKER DODGE IT?

We let the attacker try 36 plans. Against the one-round alarm, a slow drip wins every time unseen. Against the 3-round rule, **none** got through.

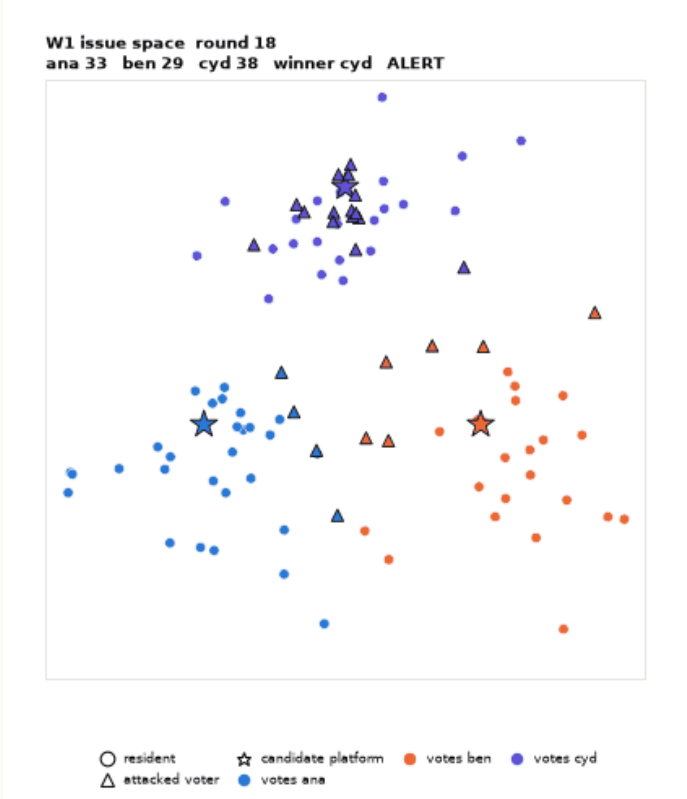
OUR REAL-WORLD READING

Monitors that only look for sudden swings miss patient manipulation. Looking a few rounds back is enough to catch it.

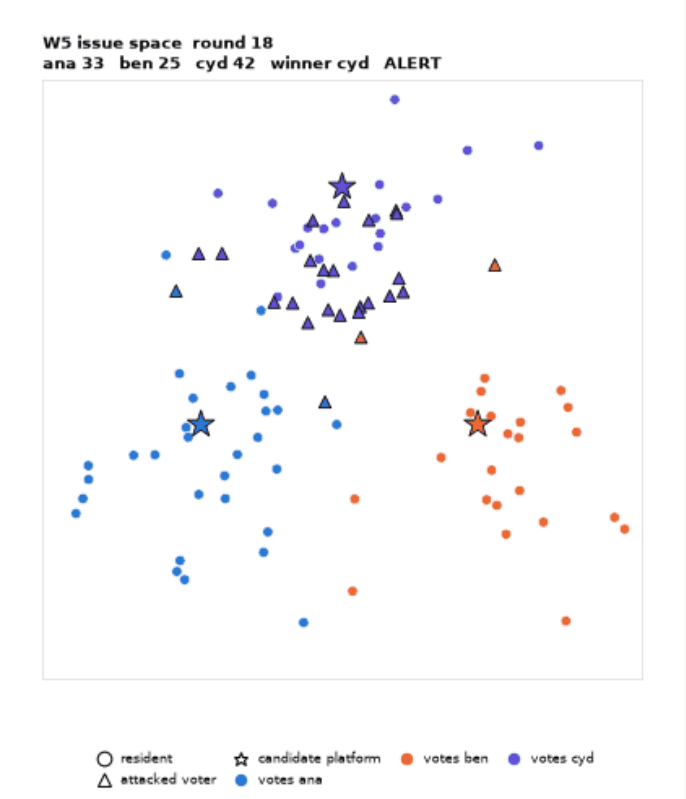
Best rule picked from 101 candidates on one set of towns, then tested on new ones · a 3-point one-round bar also works

EXPERIMENT 5 · WATCH IT HAPPEN

A blitz is loud, a slow drip is quiet



Blitz: Cyd jumps 14 points in round 1. Every alarm fires at once.

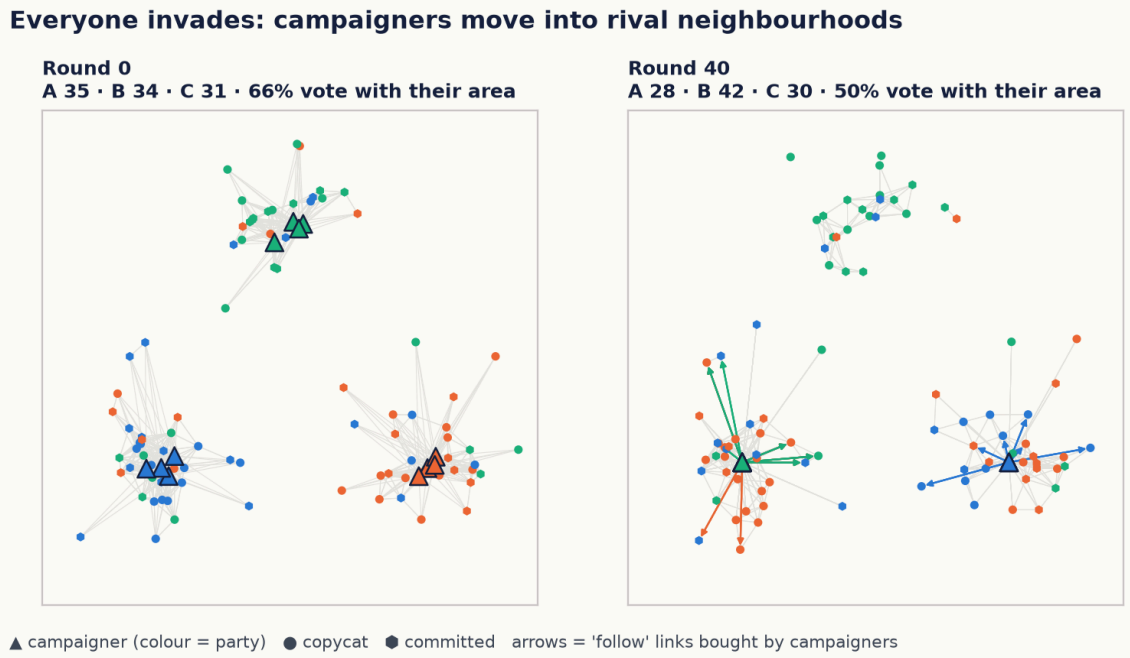


Slow drip: Cyd creeps up a few points a round and still takes the lead.

Belief space · ▲ pushed voters · ★ candidates · colour = current vote · 'ALERT' appears when the 3-round alarm fires · in the PowerPoint these play as animations

SETUP FOR EXPERIMENT 6

Campaigners who change who you hear



Everyone invades. Campaigners walk into rival neighbourhoods and get voters to follow them. By round 40 only 50% still vote with their neighbourhood.

THE TOWN

Three neighbourhoods, one per party: **A 34, B 30, C 24**. Most residents are copycats: they vote like the 4 or 5 neighbours they see, so each area becomes an **echo chamber**.

THE CAMPAIGNERS

4 per party. They move fast, can poll a wide area, and can get up to 6 voters to **follow** them, so those voters see the campaigner's vote.

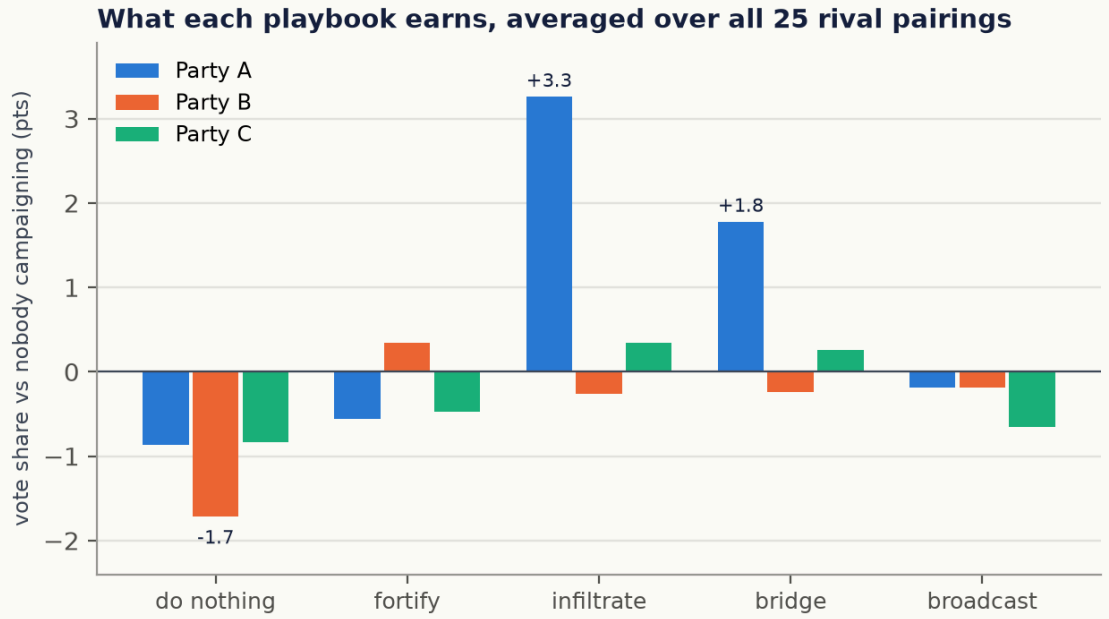
THE PLAYBOOKS

Do nothing · **defend** your home area · **invade** the biggest rival area · **work the border** · **random** followers. Every party picks one.

All 125 combinations of playbooks × 3 towns × 40 rounds · score: a party's vote share over the last 10 rounds

EXPERIMENT 6

Attack if you lead, defend if you chase



Each bar: how much a playbook earns a party on average, against every combination of its rivals' playbooks. “Fortify” is defend; “infiltrate” is invade.

WHAT WE GOT

A (biggest) should **invade**: +3.3 points. **B** should **defend**; doing nothing costs it 1.7. **C** (smallest) can't move the needle and won **0 of 375** elections.

DOES ANYONE SETTLE?

No. Each party keeps switching to beat the others, in a loop. The closest to stable is **everyone invading**.

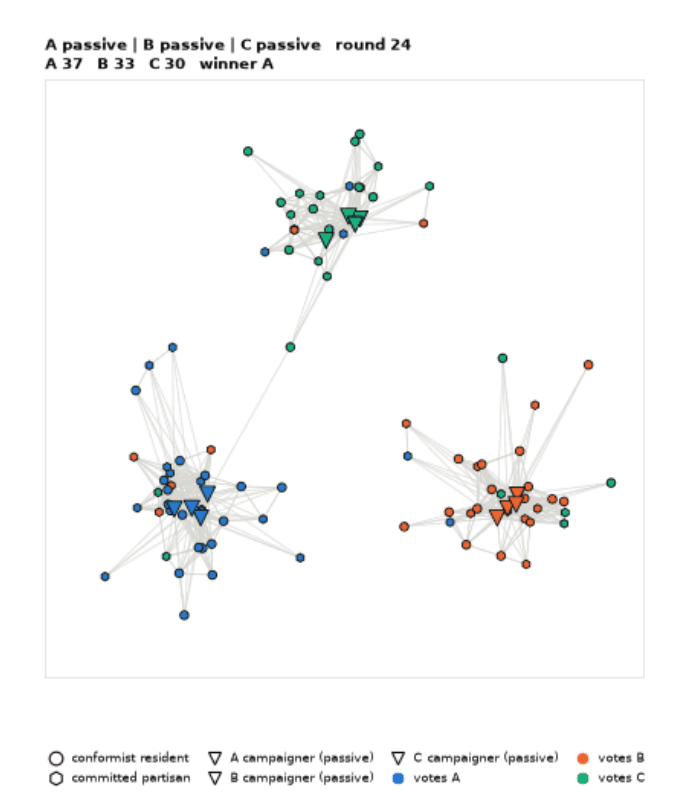
OUR REAL-WORLD READING

Campaign wars break echo chambers (79% → 55% vote with their area) but barely change who wins.

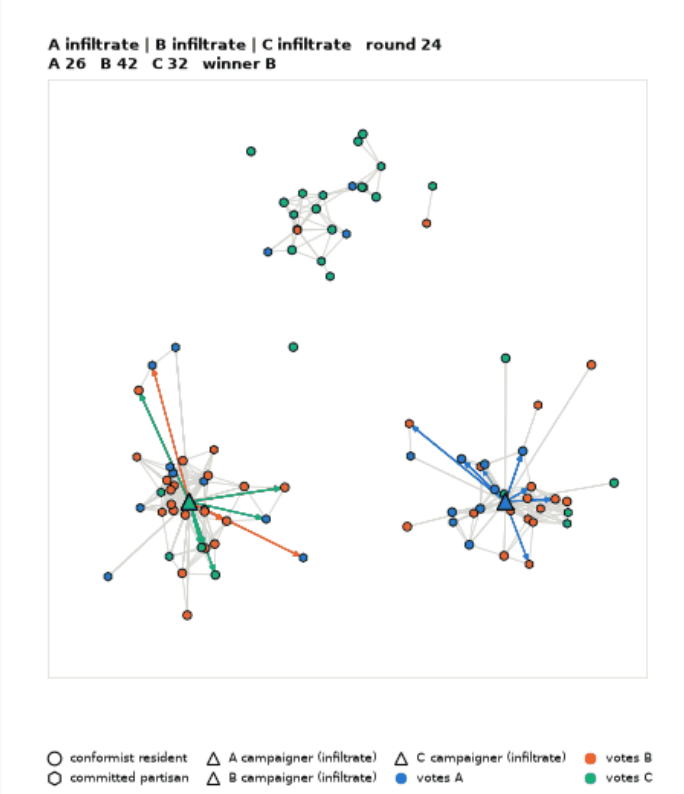
Across all 375 elections: A wins 348, B 27, C 0 · B's only upset route is working the border while A stays home

EXPERIMENT 6 · WATCH IT HAPPEN

Nobody campaigns vs everyone invades



Nobody campaigns: neighbourhoods seal themselves and A wins.



Everyone invades: follow links tear A's area open and B takes the lead.

Town map · ▲ campaigners · ● copycats · ⬢ committed · arrows = follow links bought by campaigners · in the PowerPoint these play as animations

TAKEAWAYS · PART 2

5. Watch the trend, not the jump: a 3-round look-back caught 25 of 25 slow takeovers, with no false alarms.

6. Campaign wars change who you hear, not who wins: the leader invades, the challenger defends, nobody settles, and the underdog won 0 of 375.